

Assignment 4 (User Stories and Use cases)

Project Steganography done on JPEG vs. PNG (does image compression matter)

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Stakeholder Map

Primary: Digital forensics analyst

Secondary: Steganography researcher

Hidden: Security software developer

User Stories

US-01 (primary): As a digital forensics analyst, I want to know if hidden messages are harder to detect in JPEG images than within PNG images, so that I know which images are vulnerable.

US-02 (secondary): As a image security researcher, I want to see detection results for JPEG and PNG images side by side, so that I can decide which format to use in my demonstration.

US-03 (hidden): As a security software developer, I want to know which file type hides information better, so that I can make systems check those files more closely.

Invest Self-Check

US-01

- Independent: PASS
- Negotiable: PASS
- Valuable: PASS
- Estimable: PASS
- Small: PASS
- Testable: PASS

US-02

- Independent: PASS
- Negotiable: PASS
- Valuable: PASS
- Estimable: PASS
- Small: PASS
- Testable: PASS

US-03

- Independent: PASS
- Negotiable: PASS
- Valuable: PASS
- Estimable: PASS

- Small: PASS
- Testable: PASS

Use Cases

UC-01: Compare detection for JPEG and PNG files

- Expands: US-01
- Primary actor: Digital forensics analyst
- Secondary actor: Detection tool
- Preconditions:
 - Detection tool is installed
 - 50 JPEG and 50 PNG images each have a hidden message
- Main flow:
 - i. Analyst loads the images into the tool
 - ii. Tool scans each image
 - iii. Analyst asks for the results
 - iv. Tool shows how many hidden messages it found in each format
- Alternate flow:
 - Analyst loads only one format
 - Tool shows results for that format only
- Exception flow:
 - Tool can't open an image
 - Tool skips it and lists it as an error
- Postcondition:
 - Number of hidden messages found in each format is recorded

Acceptance Criteria

AC-01.1 (Main flow)

- Given: 50 JPEG and 50 PNG images, each with a hidden message
- When: The analyst scans all 100 images
- Then: The tool shows the number of hidden messages found in each format

AC-01.2 (Exception flow)

- Given: 1 image the tool can't open
- When: The analyst scans the images
- Then: The tool skips that image and lists its file name as an error